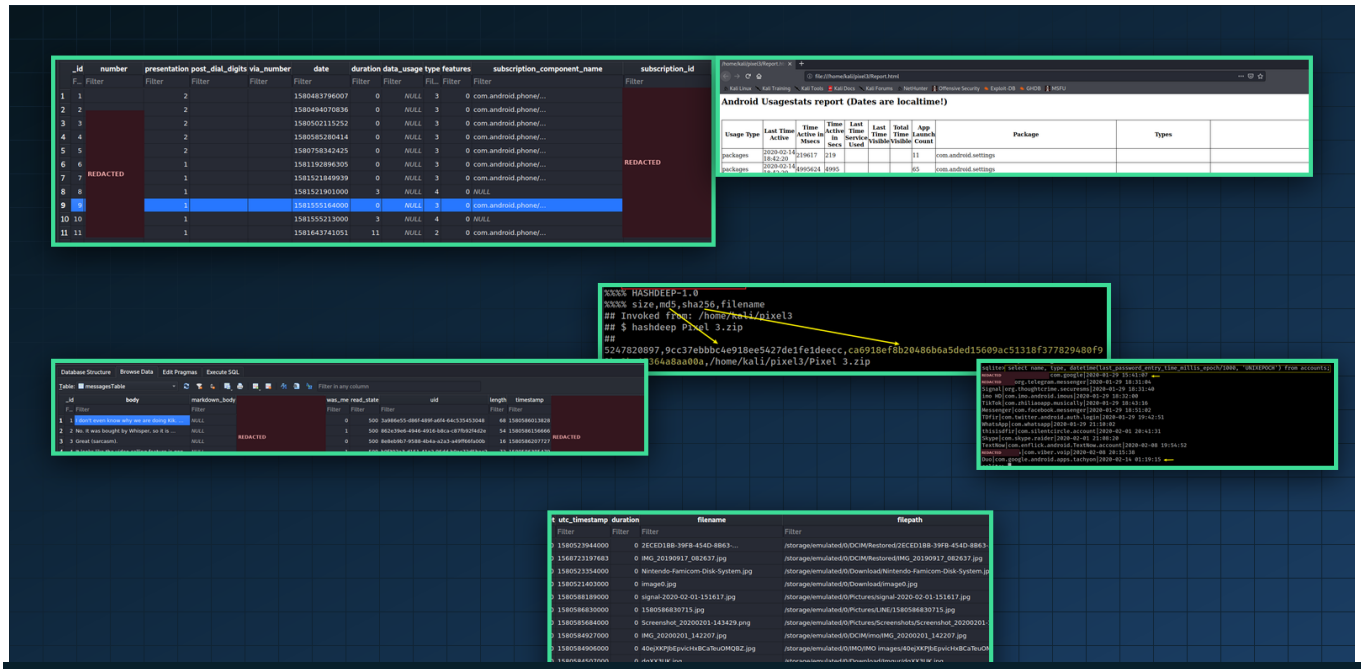


CIP-B106 | MOBILE AND IOT FORENSICS

Digital Forensic Examination Report

Google Pixel 3, Android 10 logical copy: device profile, application, communications and media findings, and event chronology

Case identifier	**CIPB106-Android10-C11/26/DFIT/17310**
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Registration number	C11/26/DFIT/17310
Cohort	Cohort 11
Submission date	October 11, 2026
Evidence examined	Pixel 3 Android 10 training logical copy (Pixel 3.zip, extracted folder "Pixel 3")

Academic Integrity Declaration

I confirm that this report reflects my own examination of the authorised course evidence and that I have identified all sources, tools, and assistance used.

Assistance used: None

Signed: Ebenezer Elikem Amankwah Ofori Dzam Date: October 11, 2026

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2 Executive Summary

This report examines the Pixel 3 Android 10 training logical copy for Case CIPB106-Android10. The archive matched the SHA-256 in the brief exactly. The handset is a Google Pixel 3 (blueline, build QQ1A.200105.003, security patch 2020-01-01) with two users, thirteen account types and two Google Fi SIM records. Play Store, package and usage records show a burst of messaging-app installs on 29 January 2020. mmssms.db and Google Messages hold mainly automated verification texts and some MMS; Kik, TextNow and WhatsApp databases hold readable messages or contacts. A 55-hour chronology (12 to 14 February 2020, UTC) links calls, voicemail, dialer look-ups, account, user, contact and messaging records, with four cross-source relationships. Principal limits: the copy is logical, acquired after TWRP root access (a Magisk folder exists, which proves nothing about the user); calendar rows, Maps databases and Photos backup state were only partly examined; some call and message direction codes are unconfirmed; findings describe device artefacts, not the person using it.

3 Evidence and Method

3.1 Evidence register and acquisition scope

The primary item is the course package Pixel 3.zip. It was retrieved with wget into ~/pixel3 on a Kali workstation, hashed with hashdeep, and unpacked to a folder named "Pixel 3" whose top level holds apex, data, metadata, mnt, product, res, sbin, system and vendor plus init scripts (Figure 1). The original ZIP was not opened for writing, renamed or re-uploaded. The brief states that the course image is a logical copy acquired after TWRP root access, not a bit-for-bit image. Data absent from a logical copy (unallocated space, deleted records not carried by the databases) cannot be assessed, and the root workflow does not show that the original user rooted the phone.

Item	Detail	Status
Pixel 3.zip	Size 5,247,820,897 bytes; working folder ~/pixel3	Hash verified (3.2)
Extracted folder "Pixel 3"	Paths below are relative to this folder, for example data/system/packages.xml	Examined read-only except where noted
Workstation tools	Kali Linux; DB Browser for SQLite, sqlite3, grep, tree, date, Firefox (for XML), git, python3	
Parser	Android-Usagestats-XML-Protobuf (usagestats_conv.py) cloned from GitHub	
Name look-up script	curl and grep against the Google Play page for each folder in data/data	

Handling statement. Databases were browsed and queried with SELECT statements. Two exports were written outside the evidence (appstate.csv and usagestats.db). Phone numbers, e-mail-format account names, serial numbers, subscriber and SIM identifiers, the Bluetooth address and coordinates are masked in the narrative and redacted in figures. No credential was recovered or disclosed.

3.2 Integrity check

hashdeep reported size 5,247,820,897, MD5 9cc37ebbbc4e918ee5427de1fe1deecc and SHA-256 ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a (Figure 2). The SHA-256 is identical, character for character, to the expected value in the brief. The brief gives no MD5, so none could be compared. The archive is therefore recorded as matching the course package.

3.3 Parser and query method

Each task started from the file system. Property files were read with cat or grep, XML in a browser, and SQLite databases in DB Browser for SQLite or sqlite3. Package names were turned into application names with a Google Play look-up script that prints the listing title for each package folder. Usage events came from the open-source usagestats parser, which converts the XML and protobuf files under data/system/usagestats into a SQLite table named data. Where the parser output was used, raw values were kept beside converted values (Appendix C).

3.4 Timestamp conversion method

Most values used here are Unix epoch milliseconds, converted by dividing by 1,000 and applying the UNIXEPOCH modifier or the date command in UTC. Examples: Play Store first_download_ms 1580317477586 gives 2020-01-29 17:04:37 UTC (checked with date -d @1580317477 -u, Figure 9); Maps last_sync_time 1581277539482000 is microseconds (divide by 1,000,000) and gives 2020-02-09 19:45:39 UTC. mmssms.db pdu dates are in seconds. Photos capture_timestamp is a local wall-clock value held as an epoch (it equals utc_timestamp plus timezone_offset).

The device time zone is America/New_York: netpolicy.xml records a billing cycle in that zone and Google Photos records a timezone_offset of -18000000 ms (UTC-5, EST) in winter 2020 and -14400000 ms (UTC-4, EDT) for a September 2019 photo. All 2020 dates in this report fall in EST, so local time is UTC minus 5 hours. All times are UTC unless labelled. The usagestats Report.html shows examination-machine local time and is not used for conversions. Whether the device clock was automatic was not examined, so times are qualified accordingly.

4 Device, Account and System Profile

4.1 Build and hardware

product/build.prop gives fingerprint google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys, build ID QQ1A.200105.003, release 10, SDK 29, device blueline, model Pixel 3, manufacturer Google, build date 17 December 2019 19:34:12 UTC (epoch 1576611252) (Figure 3). The security patch level 2020-01-01 comes from system/build.prop (Figure 4). A factory serial number appears in mnt/vendor/persist/sensors/tmd2725_als_factory.txt and in persist camera and sensor calibration files (masked here as 8CE******). This is a hardware factory value, not the Android ID or an account identifier. The Android ID in gservices.db is not reproduced.

4.2 Users and owner evidence

userlist.xml lists users 0 and 11 with nextSerialNumber 12 and deviceOwnerUserId -10000, so no device owner is declared (Figure 7). User 0 is named "This Is" (flags 19: primary, admin, initialised) and last logged in at raw 1581695517124, 2020-02-14 15:51:57 UTC. User 11 is named "User 2" (flags 16: initialised, secondary) and was created at raw 1581644358241, 2020-02-14 01:39:18 UTC, with a last login at raw 1581684601922, 2020-02-14 12:50:01 UTC. Guest restrictions block SMS, outgoing calls, Wi-Fi configuration and unknown-source installs. A scheduled Google Calendar sync job for userId 11 in data/system/job independently shows that the second user held a Google-synced profile.

4.3 Accounts

accounts_de.db lists thirteen accounts (Figure 8). The query used the UNIXEPOCH modifier, so converted times are UTC. last_password_entry_time records when credentials were last entered or the account was added, which is not the same as app installation or use. accounts_ce.db holds the same account set; its password column holds an opaque token and was not interpreted.

Account type	Last password entry (UTC)	Raw value (ms)
Google (e-mail-format name, masked)	2020-01-29 15:41:07	1580312467984
Telegram (numeric name, masked)	2020-01-29 18:31:04	1580322664981
Signal	2020-01-29 18:31:40	1580322700050
imo HD	2020-01-29 18:32:00	1580322720395
TikTok	2020-01-29 18:43:16	1580323396517
Facebook Messenger	2020-01-29 18:51:02	1580323862026
Twitter	2020-01-29 19:42:51	not captured
WhatsApp	2020-01-29 21:10:02	not captured
Silent Circle	2020-02-01 20:41:31	not captured
Skype	2020-02-01 21:08:20	not captured
TextNow	2020-02-08 19:54:52	not captured
Viber (number-form name, masked)	2020-02-08 20:15:38	not captured

Account type	Last password entry (UTC)	Raw value (ms)
Google Duo	2020-02-14 01:19:15	not captured

Table 1. accounts_de.db (data/system_de/0). Raw values not visible in the exhibits are marked "not captured"; converted values are as printed by the query.

Two Google-account-format identifiers occur in the copy: one 608 times and a second 120 times (and 43 and 36 times under data/system). The second was not attributed to a user, so no link between it and user 11 is claimed.

4.4 SIM, network and connectivity

telephony.db table siminfo holds two Google Fi records (Figure 5). One has MCC 310 and MNC 260, sim_id -1 and a number beginning 651; the other has MCC 310, MNC 120, sim_id 0 and the number +1 919 579 ****. The ICCIDs end 6693 and 3842. Subscriber identifiers differ in type: the number is a subscriber (MSISDN) value, the ICCID identifies the SIM card, and MCC and MNC identify the operator network. Carrier config files in com.android.phone give the version fi_us-12000000333.16, and netpolicy.xml records a subscriber ID (masked) with a monthly metered cycle in America/New_York. The Google Dialer voicemail_status table shows the dialer configured for the second ICCID and the Google Fi package (com.google.android.apps.tycho) with configuration state 0.

blockednumbers.db holds a single blocked number (Figure 6). Bluetooth: settings_secure.xml records bluetooth_name "Pixel 3" and a Bluetooth address (masked). Wi-Fi: WifiConfigStore.xml has an empty NetworkList; its MacAddressMap holds one entry whose network name was not legible in the exhibit. No saved network is therefore reported, and no password was recovered.

4.5 Settings and policy

locksettings.db records a lock-screen credential type of 196608 (a numeric, PIN-type quality value) with the Google trust agent enabled and the factory-reset-protection flag set. No password recovery was attempted. notification_policy.xml holds a Do Not Disturb rule set (calls and repeat callers allowed, messages blocked) and 97 package entries. gservices.db and googlesettings.db show Google service configuration, including network_location_opt_in 1 and a use-location-for-services setting; these are settings, not records of location use. GMS shared preferences record legal country US and, in Fingerprintstats and Launcherstats, background tasks last run at raw 1581560776867 and 1581560777469 (2020-02-13 02:26:16 UTC).

4.6 Acquisition and root context

data/magisk_backup_e132 and a com.topjohnwu.magisk data folder are present in the copy, consistent with the TWRP and Magisk workflow the brief describes. This is acquisition provenance. It is not evidence that the original user rooted, modified or compromised the phone, and no user-activity finding depends on it.

4.7 Installed applications and install evidence

data/data holds about one hundred package folders. By package prefix, com.android.* and the providers are AOSP, com.google.* is Google Mobile Services, and the rest are third party (Discord, TextNow, Facebook Messenger, Imgur, imo, Instagram, Threads, MeWe, Wickr, Cyberdust, SilentPhone, Skout, Skype, Snapchat, Spotify, Gallery Vault, Magisk, Twitter,

Venmo, Viber, WhatsApp, Wire, TikTok, LINE, Kik and others). The Play Store look-up and a keyword scan for "money" flagged TextNow, Messenger, Google Wallet and Venmo as money-related listings. Lookup unmatched results. packages.list gives UID 10141 for com.google.android.gms and 10187 for com.google.android.gm; packages.xml for Gmail gives codePath under /data/app, version 62049238 and installer com.android.vending.

com.android.vending localappstate.db (table appstate) records install activity with first_download_ms: Data Transfer Tool 1580312356256 (2020-01-29 15:39:16 UTC); WeChat 1580317477586 (17:04:37); LINE 1580317512405 (17:05:12); WhatsApp 1580317744179 (17:09:04); Twitter 1580317762141 (17:09:22); Snapchat 1580317779105 (17:09:39); Imgur 1580317923118 (17:12:03). Account on each row is masked. library.db gives WhatsApp ownership purchase_time 1543518467665, 2018-11-29 19:07:47 UTC, so the account held the app before this device was set up; it does not show use on this phone. Snapchat's own preference INSTALL_ON_DEVICE_TIMESTAMP (raw 1580317793035) is 17:09:53 UTC, 14 seconds after the Play Store download record; the two independent records agree.

package-usage.list, package-dcl.list, package-watchdog.xml and the usagestats folder (users 0 and 11; daily files from 1580917965671 to 1581696001773, plus weekly, monthly and yearly) provide usage evidence. The parser output (Figure 10, Figure 11) includes LINE events on 9 February 2020: MainActivity foreground at 17:49:24 UTC (raw 1581270564168 onward), ChatHistoryActivity foreground at 18:05:20 UTC (raw 1581271520743), SelectLocationActivity stopped at 18:05:21, then Viber HomeActivity foreground at 18:05:51 (raw 1581271551366) and ConversationActivity at 18:05:52. These show application lifecycle events only; the user, message content and any location shared are not shown.

Cross-source relationship (Task B). packages.list assigns UID 10141 to com.google.android.gms. netpolicy.xml carries uid-policy entries for 10141 and for 1110141, the same app under user 11 (user ID x 100,000 + app UID). Package metadata and a network policy file therefore agree on the app and independently confirm that user 11 had its own GMS instance. A second pair is the Snapchat install match above.

5 Artefact Findings

5.1 AOSP Telephony Provider (mmssms.db)

data/data/com.android.providers.telephony/databases/mmssms.db has 18 tables including sms, pdu, part, addr, canonical_addresses and threads. The sms table (Figure 12, Figure 13) holds messages whose creator is com.google.android.apps.messaging and whose seen flag is 1. These are short-code and long-number texts carrying verification codes (codes omitted here): LINE at 1580322045251 (2020-01-29 18:20:45 UTC), Telegram at 1580322646631 (18:30:46), Signal at 1580322696889 (18:31:36), imo at 1580322728551 (18:32:08), two Skout texts (18:33:40 and 18:35:03), Facebook (18:50:13), an Imgur code at 1580328970337 (20:16:10), and a Dust code at 1580332533357 (21:15:33). Later threads include Wickr (2020-02-01 01:00:03 UTC) and Wire (01:04:13). Direction is inferred from content (service-originated), since the type column was not captured, and date_sent runs one to two seconds before date on each row.

MMS evidence sits in pdu and part. Thread 17 holds pdu rows (dates in seconds) 1580587493 (2020-02-01 20:04:53 UTC, msg_box 2, m_type 128), 1580587564 (20:06:04, msg_box 1, m_type 132), 1580587672 (20:07:52) and 1580587692 (20:08:12). The part table links to each pdu by mid and holds SMIL layout parts plus text parts such as "Here's Signal." and a longer text beginning "It looks like the messaging is run through the". Because each pdu has an SMIL part and a text part, the four pdu rows are four messages, not eight. The threads table records thread 13 updated at 1581628344000 (2020-02-13 21:12:24 UTC) with a notification-style snippet, thread 17 updated at 1581685808000 (2020-02-14 13:10:08 UTC), and three empty threads (18 to 20) at 15:00:47 to 15:00:53 UTC on 14 February.

Google Messages bugle_db presents the same records. Its messages table carries sms_message_uri values such as content://sms/2 and content://sms/3, and its conversations table holds sms_thread_id 4 for the LINE conversation, matching mmssms thread 4 (). Message protocol 0 indicates SMS. The linked entries are copies of the provider records and are not counted as extra messages. RCS context was not examined. The participants table includes the device's own number with a profile label.

5.2 Contacts, calls and voicemail

contacts2.db accounts lists Google, Telegram, Silent Circle, Signal, Skype, Viber, WhatsApp and Duo. Contact _id 10 (display name masked) aggregates eight raw contacts, one per account, with a phone number in the 919-area code, times_contacted 0 and contact_last_updated_timestamp 1581684660833 (2020-02-14 12:51:00 UTC). An address-book label shows how the user or an account stored a number. It does not show who used that number.

calllog.db table calls holds 14 rows (Figure 14, Figure 15). Type values follow the Android convention: 1 incoming, 2 outgoing, 3 missed and 4 voicemail. Rows 1 to 5 are withheld-number (presentation 2) missed calls from 2020-01-31 15:16:36 UTC to 2020-02-02, on the first SIM. Row 6 is a missed call from a +33 number at 2020-02-08 20:14:56 UTC. Row 7 is a missed call at 1581521849939 (2020-02-12 15:37:29 UTC) and row 8 is a 3-second voicemail record at 1581521901000 (15:38:21), both with the same number as the blocked-number entry. Row 9 is a missed call at 1581555164000 (2020-02-13 00:52:44 UTC) from a number geocoded to Questa, NM, and row 10 is a 3-second voicemail record at 1581555213000 (00:53:33). Rows 8 and 10 carry voicemail URIs content://com.android.voicemail/voicemail/8 and /10 with source package com.google.android.apps.tycho (Google Fi). Row 11 is an outgoing call (type 2) of 11 seconds at

1581643741051 (2020-02-14 01:29:01 UTC) to a number saved under contact 10. Rows 12 to 14 are calls of 65, 86 and 91 seconds at 01:30:24, 13:05:57 and 14:41:18 UTC on 14 February. The type of rows 12 to 14 was not captured.

Google Dialer annotated_call_log.db (Figure 16) repeats the same timestamps and durations for rows 1 to 7, 9 and 11 to 14, but it has no rows with _id 8 or 10. The two voicemail entries therefore exist only in the AOSP call log. Voicemail audio files exist in the Contacts provider files folder (two voicemail directories). Dialer voicemail.db has an empty greeting table (), which is a supported finding that no greeting is stored. Dialer phone_lookup_history.db records look-ups at 1581539995439 (2020-02-12 15:39:55 UTC), 1581555192770 (2020-02-13 00:53:12), 1581643892679 (2020-02-14 01:31:32), and earlier at 1580483823810 and 1581192904815 (Figure 17). suggest_contact_database holds a ContactScore of 1.0 for the saved contact's number, and smartdial_table updated that contact at 1581643892894 (01:31:32 UTC).

5.3 Calendar

calendar.db (data/data/com.android.providers.calendar/databases) has 15 tables, including Attendees, CalendarAlerts, Calendars and Events. Row counts were not captured, so neither events nor an empty calendar is reported. A periodic sync job for com.google.android.calendar under user 11 exists, which shows sync configuration, not events.

5.4 Google Maps and Google Photos

Maps databases are in data/data/com.google.android.apps.maps/databases (gmm_myplaces.db, gmm_storage.db, gmm_sync.db, ugc_photos_location_data.db). gmm_sync.db sync_item_data holds three corpus-11 rows with latitude_e6 and longitude_e6 values (Figure 18); in E6 microdegrees, 35734602 means 35.734602 degrees. The coordinates are masked here and were not looked up on any web service. One row has timestamp 1554407343604 (2019-04-04 19:49:03 UTC). Three corpus-13 rows have client_id prefixes of 1581273852322 (2020-02-09 18:44:12 UTC), and the sync_corpus table records last_sync_time 1581277539482000 microseconds (2020-02-09 19:45:39 UTC). files/offline_saved_directions.data contains turn-by-turn text for a saved route (West Elm Avenue and North Main Street). gmm_storage_table has CompletedNavigation and CommuteSetupNotificationModel keys whose blobs were not decoded. The photos_top_feature table is empty. These records show saved or synchronised Maps data. They do not show that the phone was at those points.

Google Photos gphotos-1.db has 109 tables. The media table (Figure 19) stores utc_timestamp, timezone_offset (-18000000 for winter items) and capture_timestamp, which equals the UTC time minus five hours. local_media (Figure 20) lists files with paths, including DCIM/Restored items, Download, Pictures/Screenshots, Pictures/LINE, IMO and Imgur folders. Two entries test the time model: IMG_20190917_082637.jpg has utc_timestamp 1568723197683 (2019-09-17 12:26:37 UTC) and an offset of -14400000 ms, which gives 08:26:37 local, matching the time in the file name; IMG_20200201_142207.jpg in the imo folder has utc_timestamp 1580584927000 (2020-02-01 19:22:07 UTC), which gives 14:22:07 EST, again matching its name. signal-2020-02-01-151617.jpg (20:16:29 UTC, 15:16:29 EST) and Screenshot_20200201-143429.png (19:34:44 UTC, 14:34:44 EST) differ from their names by 12 and 15 seconds, consistent with save time rather than creation time. Backup state (backup_manager.xml and backup tables) was not read.

5.5 Third-party applications

Kik (kik.android). Version 15.19.0.22104 (from the app's analytics file). kikCoreDatabase.db CoreTable shows username ThisIsDFIR active, KikPreferences shows first name "ThisIs" and last name "DFIR", and the e-mail confirmed flag is false. Google measurement preferences give first_open_time 1580322534533 (2020-01-29 18:28:54 UTC) and the login time is 1580322563292 (18:29:23). kikDatabase.db messagesTable (Figure 21) holds six messages with one partner: ids 1 to 4 have text and ids 5 and 6 have no body (length 0), consistent with media items (KikContentTable lists a "Gallery" item of 90,417 bytes with preview thumbnails). was_me 0 is received and 1 is sent. The times are 2020-02-01 19:40:13, 19:42:36, 19:43:27, 19:46:25, 19:47:04 and 19:48:04 UTC (raw 1580586013828 to 1580586484810). The text concerns Kik's ownership and the loss of its video-calling feature. The media thumbnails were not opened.

TextNow (com.enflick.android.TextNow). Preferences record a TextNow phone number (masked, 984 area code) and username "thisisdfir". textnow_data.db has 13 tables including contacts, conversations and messages. The messages table (Figure 22) shows an exchange with one number at raw 1581192046000 (2020-02-08 20:00:46 UTC) "Hi there!", 20:01:27 "Back at it again, I see.", 20:03:06 "Yep! Have to keep generating data.", then rows with message_type 102 and 201 at 1581192341146 (20:05:41), and a picture message with an attachment path under Pictures/Twitter at 1581192625000 (20:10:25). The meaning of message_direction (1 and 2) was not confirmed. The text of the exchange suggests that value 1 may mean sent, but the usual TextNow convention is the reverse, so no direction is asserted. The TextNow account entry (19:54:52 UTC) precedes this exchange by six minutes.

WhatsApp (com.whatsapp). registration.RegisterPhone.xml records country code 1 and the registered number (masked). The Play Store record puts install at 2020-01-29 17:09:04 UTC and the account at 21:10:02 UTC. wa.db wa_contacts (Figure 23) has three rows: status@broadcast, a vendor-style number whose status date is 1556918335000 (2019-05-03 21:18:55 UTC), and the same 919 number held by contact 10 (status_timestamp 1581195270000, 2020-02-08 20:54:30 UTC). msgstore.db (106 tables) has a messages table with status-broadcast rows dated 1487100001000 onward, which are placeholders predating the install, and incoming media rows with received_timestamp 1580323038260 (2020-01-29 18:37:18 UTC). A join on chat 3 returns three texts: "Hi there! I switched over.", "Awesome!" and "The Osbourne!". Their direction and times were not captured.

Other applications. Snapchat (com.snapchat.android) holds an INSTALL_ON_DEVICE_TIMESTAMP of 17:09:53 UTC on 29 January and a LONG_CLIENT_ID_DEVICE_TIMESTAMP of raw 1581635948484 (2020-02-13 23:19:08 UTC); a profile preference has key_timestamp 1581641453509 (2020-02-14 00:50:53 UTC) and the profile display name. Its databases (core.db, main.db, memories.db, arroyo.db and others) were listed only. Signal, Wickr, Wire, Telegram and imo have account and SMS evidence (5.1, 4.3) but their content databases were not examined.

5.6 Second burst of activity (1 February 2020)

Outside the chronology window, 1 February 2020 shows a connected sequence across five artefact families: imo image saved at 19:22:07 UTC, screenshot at 19:34:44, Kik messages from 19:40:13 to 19:48:04, MMS pdu rows from 20:04:53 to 20:08:12, a LINE picture at 19:53:50, a Signal image at 20:16:29, and the Silent Circle (20:41:31) and Skype (21:08:20) accounts. This shows that several messaging applications were being set up and tested on the device in one evening. It does not identify the person who used them.

6 Event Chronology

Window and justification. The window runs from 2020-02-12 15:00 UTC to 2020-02-14 22:30 UTC (55.5 hours; 10:00 on 12 February to 17:30 on 14 February EST). It holds the densest run of records across calls and voicemail, dialer look-ups, accounts and users, contacts, SMS threads and an application preference, six families in all, and includes both user creation and a continuous group of calls. Times are UTC and EST (UTC-5), the device zone being America/New_York (Section 3.4). Raw values are Unix milliseconds unless stated.

#	UTC (converted)	EST	Raw value	Source / table	Event	Confidence
E1	2020-02-12 15:37:29	10:37:29	1581521849939	calllog.db / calls id 7; annotated_call_log.db id 7	Missed call (type 3, 0 s) from a number that is also in blockednumbers.db	High for time; block date unknown
E2	2020-02-12 15:38:21	10:38:21	1581521901000	calllog.db / calls id 8 (type 4)	Voicemail record (3 s), URI voicemail/8, source Google Fi package	High; absent from annotated_call_log.db
E3	2020-02-12 15:39:55	10:39:55	1581539995439	phone_lookup_history.db	Dialer looked up the same number	High; shows dialer processing, not user action
E4	2020-02-13 00:52:44	19:52:44 (12 Feb)	1581555164000	calllog.db / calls id 9; annotated id 9	Missed call; geocoded to Questa, NM	High for time; location is number region only
E5	2020-02-13 00:53:12	19:53:12 (12 Feb)	1581555192770	phone_lookup_history.db	Dialer look-up of that number	High
E6	2020-02-13 00:53:33	19:53:33 (12 Feb)	1581555213000	calllog.db / calls id 10 (type 4)	Voicemail record (3 s), URI voicemail/10	High; absent from annotated log
E7	2020-02-13 21:12:24	16:12:24	1581628344000	mmssms.db / threads id 13	Thread 13 date updated; snippet is a platform notification text	Moderate: thread-level value only
E8	2020-02-13 23:19:08	18:19:08	1581635948484	Snapchat shared_prefs (LONG_CLIENT_ID_DEVICE_TIMESTAMP)	App client-ID timestamp recorded	Moderate; meaning of key not verified

#	UTC (converted)	EST	Raw value	Source / table	Event	Confidence
E9	2020-02-14 01:19:15	20:19:15 (13 Feb)	not captured	accounts_de.db / accounts (Duo)	Google Duo account entry	High for time; entry time only
E10	2020-02-14 01:29:01	20:29:01 (13 Feb)	1581643741051	calllog.db id 11; annotated id 11	Outgoing call, 11 s, to the number saved under contact 10	High; see R1
E11	2020-02-14 01:30:24	20:30:24 (13 Feb)	1581643824061	annotated_call_log.db id 12	Call, 65 s, same number	High for time; type not captured
E12	2020-02-14 01:31:32	20:31:32 (13 Feb)	1581643892679 / 1581643892894	phone_lookup_history.db; smartdial_table	Look-up and smart-dial update for that number and contact	High; see R1
E13	2020-02-14 01:39:18	20:39:18 (13 Feb)	1581644358241	system/users/11.xml	User 11 ("User 2") created	High
E14	2020-02-14 12:50:01	07:50:01	1581684601922	system/users/11.xml	User 11 last logged in	High
E15	2020-02-14 12:51:00	07:51:00	1581684660833	contacts2.db / contacts id 10	Contact record last updated	High; updater unknown
E16	2020-02-14 13:05:57 / 13:10:08	08:05:57 / 08:10:08	1581685557104 / 1581685808000	calllog.db id 13; mmssms.db threads id 17	Call of 86 s, then thread 17 updated four minutes later	High for times; link between them is context only
E17	2020-02-14 14:41:18	09:41:18	1581691278351	calllog.db / annotated id 14	Call, 91 s, same number	High for time; type not captured
E18	2020-02-14 15:00:47 to 15:00:53	10:00:47 to 10:00:53	1581692447000 to 1581692453000	mmssms.db / threads ids 18 to 20	Three empty threads created	Moderate: NULL snippets
E19	2020-02-14 15:51:57	10:51:57	1581695517124	system/users/0.xml	User 0 last logged in	High

Table 2. Event chronology, 12 to 14 February 2020. Raw values are as stored in the source; E16 and E18 list two or three values.

Corroborated relationships

Ref	Relationship	Independent sources	Strength and qualification
R1	The outgoing call at 01:29:01 UTC, the call at 01:30:24, the look-up and smart-dial update at 01:31:32, and a saved contact with that number (contact 10, 919 area code) agree on number and time order (E10 to E12). The same number is stored for contact 10 in wa.db.	calllog.db, annotated_call_log.db, phone_lookup_history.db, smartdial_table, contacts2.db, wa.db	Strong: five sources, two apps. The saved label does not prove who used the number.
R2	The number blocked in blockednumbers.db also appears in a missed call and a voicemail record at 15:37:29 and 15:38:21 UTC, with a dialer look-up at 15:39:55 (E1 to E3).	blockednumbers.db, calllog.db, annotated_call_log.db, phone_lookup_history.db	Moderate: the date of blocking is not recorded, so a block in force at call time is not shown.
R3	The Duo account at 01:19:15 UTC is supported by a Duo account in contacts2.db, an AccountAuthenticator entry for com.google.android.apps.tachyon (UID 10209) and a package folder.	accounts_de.db, contacts2.db, registered_services, data/data	Strong for account existence; entry time is not first use.
R4	User 11 existed on 14 February: users/11.xml (E13, E14), job records for userId 11, and netpolicy UID 1110141.	users xml, data/system/job, netpolicy.xml	Strong for existence; who used the profile is not shown.

7 Interpretation and Limits

7.1 What the evidence supports

- The package matches the course archive on SHA-256, and the device is a Pixel 3 on Android 10 (SDK 29), patch 2020-01-01, in the America/New_York zone.
- Two users and thirteen accounts exist, and a group of messaging apps was installed and signed in on 29 January 2020.
- Calls, voicemail records, dialer look-ups, contacts, users and accounts agree on a short sequence on 12 to 14 February 2020 (Section 6).
- Kik, TextNow and WhatsApp hold readable communication or contact records attributed to those applications.

7.2 What remains uncertain

- Calendar rows, Maps saved places, Photos backup and deleted state, RCS content and the Snapchat, Signal, Wickr, Wire and imo databases are not established.
- Call type for rows 12 to 14, message direction in TextNow and WhatsApp, and the WhatsApp message times were not captured.
- The parser-to-source comparison for usagestats was only partly completed (Appendix E).

7.3 Contradictions and alternative explanations

- The annotated dialer log lacks the two voicemail rows that the AOSP log holds. Either the dialer does not annotate voicemail entries or they were not synchronised. The AOSP voicemail URIs show the entries are real.
- The imo account (18:32:00 UTC) is dated eight seconds before its code text (18:32:08), while Signal (18:31:40) follows its text (18:31:36) by four seconds and Telegram (18:31:04) follows its text (18:30:46) by eighteen. Account times record credential entry, not code delivery, so a small difference either way is expected.
- The call-log phone_account_address for some rows is a number that differs from the SIM number in telephony.db. Google Fi can use more than one number, so this was not resolved.
- The WhatsApp ownership record (2018) predates this phone. It shows prior purchase on the Google account, not prior use of the app on this device.
- A device record does not identify a person. Calls, messages and photos could involve other users, other devices syncing to the account, or the training workflow itself. The TextNow text "Have to keep generating data" shows that the examined data was created for training.

7.4 Four evidence-based findings

Finding 1: device and account context. A Pixel 3 (blueline) on Android 10, patch 2020-01-01, with users 0 and 11 and thirteen accounts, two Google Fi SIM records, one blocked number and a PIN-type screen lock. Limit: a logical copy acquired after TWRP root access; no owner is declared; Wi-Fi networks and the SSID were not read.

Finding 2: system and application use. Play Store and Snapchat records agree on installs on 29 January 2020, and usagestats shows LINE and Viber lifecycle events on 9 February. Limit: usage events do not show who operated the apps or what was said; parser validation is partial.

Finding 3: communications and organiser activity. Most SMS are automated verification texts; MMS and Google Messages records link to the same threads; calls and voicemail records form the sequence in R1 and R2. Limit: call and message direction gaps, no calendar rows, and a saved contact is not proof of identity.

Finding 4: reconstructed chronology. Nineteen events from six artefact families over 55.5 hours, with four relationships across independent sources. Limit: several rows are thread-level or entry-time values, and two outcomes (R2, E7) are moderate.

Closing statement. This report describes what the artefacts record. It does not decide whether any person committed an offence and does not identify the person who used the device.

Appendix A Finding Log

The format-example row in the brief has been removed. Every row retains source, raw value, converted time and qualification.

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-01	Archive matches the brief's SHA-256; MD5 recorded	Pixel 3.zip (hashdeep)	ca6918ef...a00a; 9cc37ebb...eecc	Not applicable	MD5 not given in brief
F-02	Pixel 3, blueline, Android 10, SDK 29, build QQ1A.200105.003	product/build.prop	1576611252 (build date)	2019-12-17 19:34:12 UTC	Property values only
F-03	Security patch 2020-01-01	system/build.prop	ro.build.version.security_patch	Not applicable	Patch level, not update date
F-04	Two users: 0 ("This Is") and 11 ("User 2")	system/users/userlist.xml, 0.xml, 11.xml	1581695517124; 1581644358241; 1581684601922	2020-02-14 15:51:57; 01:39:18; 12:50:01 UTC	No device owner declared
F-05	Thirteen accounts	system_de/0/accounts_de.db	1580312467984 to 1580323862026 (six captured)	2020-01-29 15:41:07 to 2020-02-14 01:19:15 UTC	Entry time, not first use

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-06	Two Google Fi SIM records, MCC 310 (MNC 260, 120)	user_de/0/com.android.providers.telephony/databases/telephony.db / siminfo	Not applicable	Not applicable	ICCIDs and numbers masked
F-07	One blocked number	user_de/0/com.android.providers.blockednumber/databases/blockednumbers.db	Not applicable	Not applicable	Block date not recorded
F-08	Bluetooth name Pixel 3; empty Wi-Fi NetworkList	system/users/0/settings_secure.xml; misc/wifi/WifiConfigStore.xml	Not applicable	Not applicable	SSID not read
F-09	PIN-type lock credential; trust agent enabled	system/locksettings.db	196608	Not applicable	No recovery attempted
F-10	Magisk folder in copy	data/magisk_backup_e132; data/data/com.topjohnwu.magisk	Not applicable	Not applicable	Acquisition provenance; not user rooting
F-11	Messaging-app installs on 29 Jan 2020	com.android.vending/databases/localappstate.db / appstate	1580317477586 (WeChat) to 1580317923118	2020-01-29 17:04:37 to 17:12:03 UTC	Download time; manual date check matched
F-12	Snapchat install time matches Play record	snapchat shared_prefs; localappstate.db	1580317793035; 1580317779105	17:09:53; 17:09:39 UTC	Two sources agree within 14 s

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-13	LINE and Viber lifecycle events	system/usagestats/0/daily (parser: usagestats.db / data)	1581271520743; 1581271551366	2020-02-09 18:05:20; 18:05:51 UTC	Parser output; source files not re-read
F-14	Verification and alert SMS (no content recovered beyond text)	mmssms.db / sms ids 2 to 15	1580322045251 to 1580518803531	2020-01-29 18:20:45 to 2020-02-01 01:00:03 UTC	Direction inferred
F-15	Four MMS in thread 17	mmssms.db / pdu, part	1580587493 to 1580587692 (s)	2020-02-01 20:04:53 to 20:08:12 UTC	SMIL and text parts counted once
F-16	Fourteen call records; two voicemail records	calllog.db / calls	1580483796007 to 1581691278351	2020-01-31 15:16:36 to 2020-02-14 14:41:18 UTC	Type of rows 12-14 not captured
F-17	Dialer log omits voicemail rows 8 and 10	annotated_call_log.db / AnnotatedCallLog	Not applicable	Not applicable	Compare with calllog.db

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-18	Look-ups and smart-dial update for saved number	phone_lookup_history.db; smartdial_table	1581643892679; 1581643892894	2020-02-14 01:31:32 UTC	Dialer processing, not user action
F-19	Contact 10 with eight raw contacts	contacts2.db / contacts, raw_contacts	1581684660833	2020-02-14 12:51:00 UTC	Label is not identity
F-20	Saved Maps rows (E6 coordinates masked) and offline directions	maps/databases/gmm_sync.db; files/offline_saved_directions.data	1554407343604; 1581277539482000 (us)	2019-04-04 19:49:03; 2020-02-09 19:45:39 UTC	Not evidence of presence
F-21	Photos time model confirmed by file names	photos/databases/gphotos-1.db / media, local_media	1568723197683; 1580584927000	2019-09-17 12:26:37; 2020-02-01 19:22:07 UTC	Offsets -14400000 and -18000000
F-22	Kik: six messages with one partner	kik.android/databases/*.kikDatabase.db / messagesTable	1580586013828 to 1580586484810	2020-02-01 19:40:13 to 19:48:04 UTC	Two rows have no body

ID	Finding	Source	Raw time or value	Converted time	Qualification
F-23	TextNow exchange with one number	TextNow/databases/textnow_data.db / messages	1581192046000 to 1581192625000	2020-02-08 20:00:46 to 20:10:25 UTC	Direction coding unconfirmed
F-24	WhatsApp contacts and chat texts	whatsapp/databases/wa.db, msgstore.db	1581195270000; 1580323038260	2020-02-08 20:54:30; 2020-01-29 18:37:18 UTC	Times and direction not captured

Appendix B Screenshots with Captions

Figures are the examiner's own screenshots from the Kali workstation. Each caption names the source, database or file shown. Phone numbers, account names, ICCIDs and coordinates are redacted in Figures 5, 6, 8, 12, 14, 15, 16, 17, 18, 21, 22, 23.



Figure 1. *Retrieval of Pixel 3.zip with wget into ~/pixel3 on the Kali workstation; ls -l shows 5,247,820,897 bytes (the date column shows "Nov 21 22:14", year not displayed).*

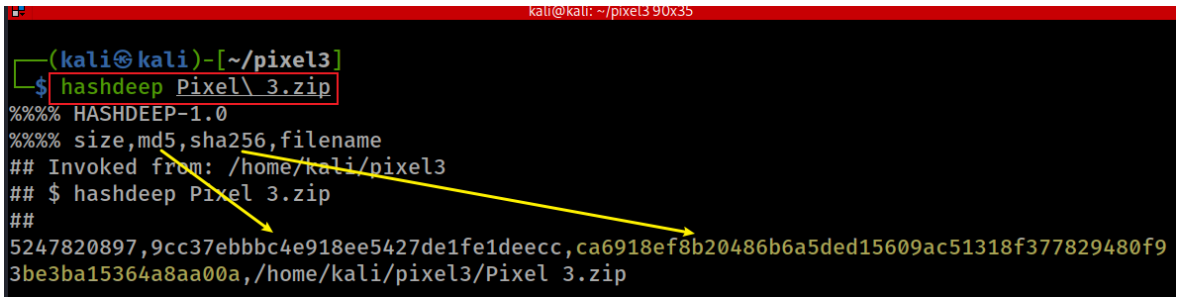


Figure 2. *hashdeep output for Pixel 3.zip: size, MD5 and SHA-256 (Section 3.2). The SHA-256 matches the value printed in the brief.*

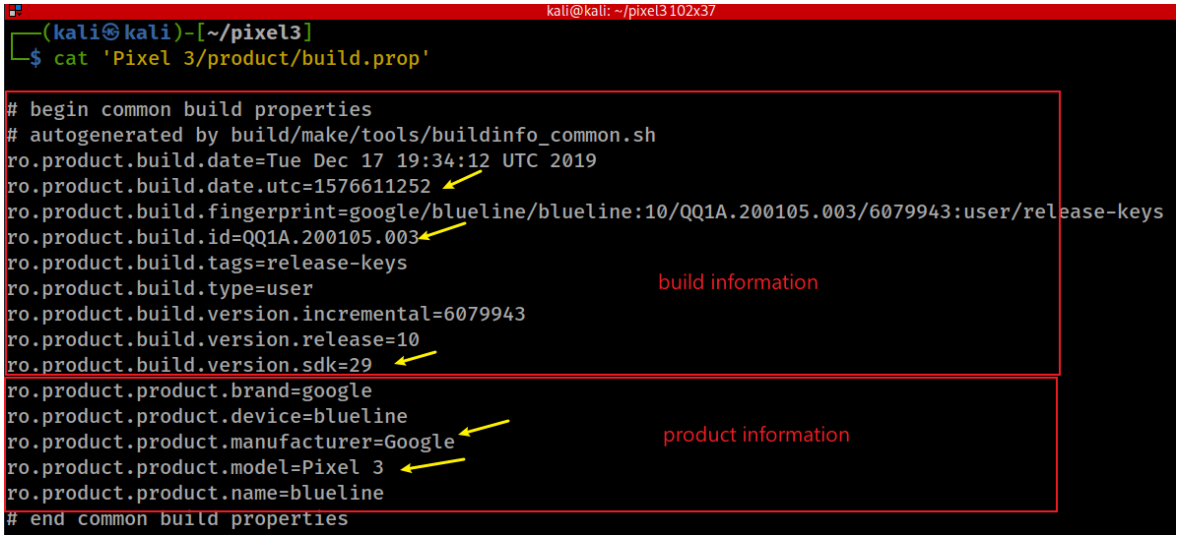


Figure 3. *cat of Pixel 3/product/build.prop: fingerprint, build ID QQ1A.200105.003, SDK 29, device blueline, model Pixel 3, build date 17 Dec 2019 (epoch 1576611252).*

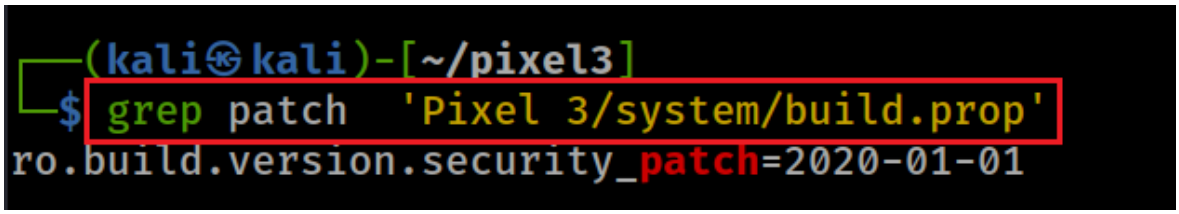


Figure 4. *grep patch on Pixel 3/system/build.prop: ro.build.version.security_patch=2020-01-01.*

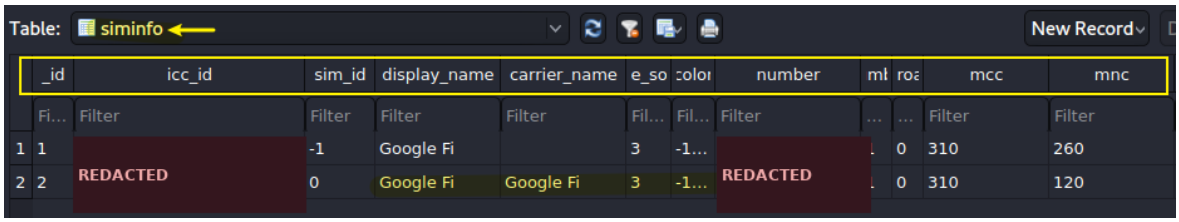


Figure 5. *telephony.db, table siminfo: two Google Fi SIM records with MCC 310 and MNC 260 / 120. ICCIDs and subscriber numbers are redacted.*

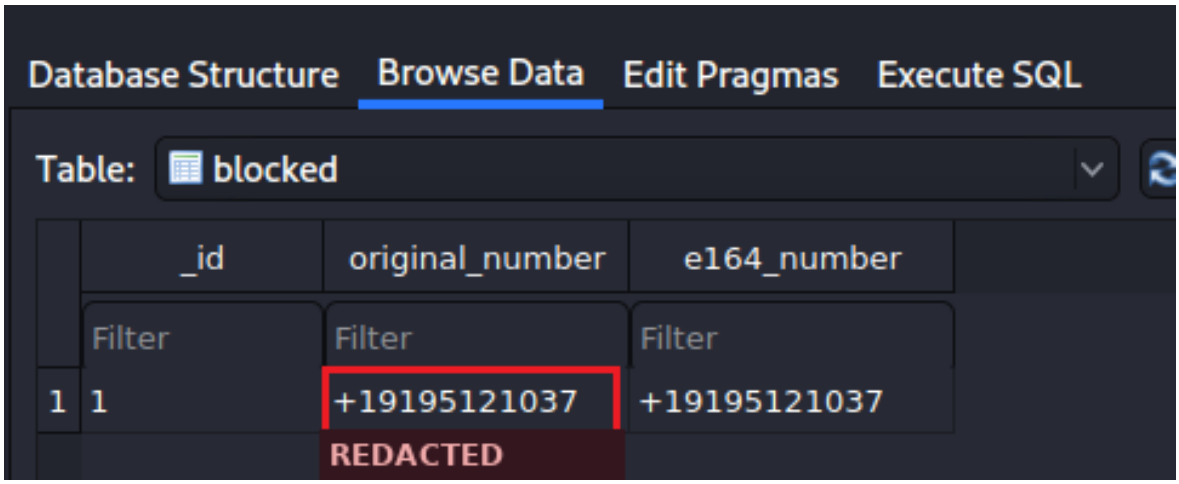


Figure 6. *blockednumbers.db, table blocked: one blocked number (value in the red box redacted in this figure; the same number appears in Table 6).*



Figure 7. *users/userlist.xml entries via 0.xml and 11.xml: user 0 "This Is" (flags 19) and user 11 "User 2" (flags 16), with created and lastLoggedIn values in epoch milliseconds.*

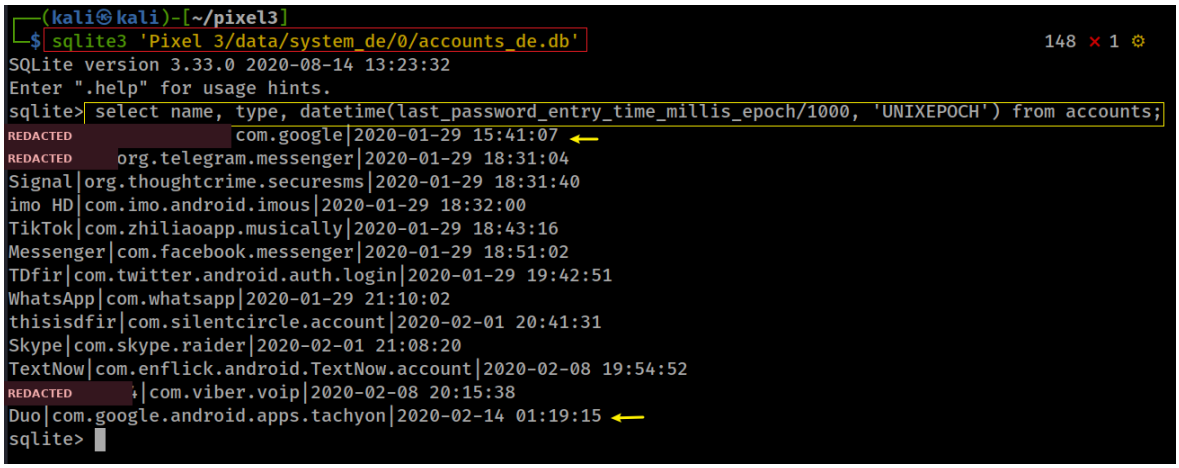


Figure 8. *accounts_de.db query converting last_password_entry_time_millis_epoch with the UNIXEPOCH modifier (UTC). Google, Telegram and Viber account names are redacted.*

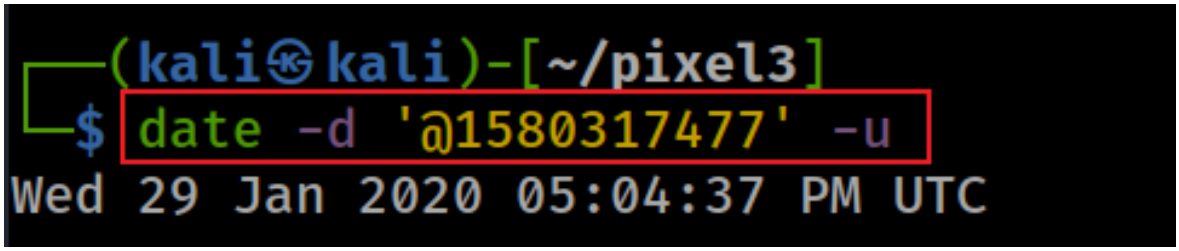
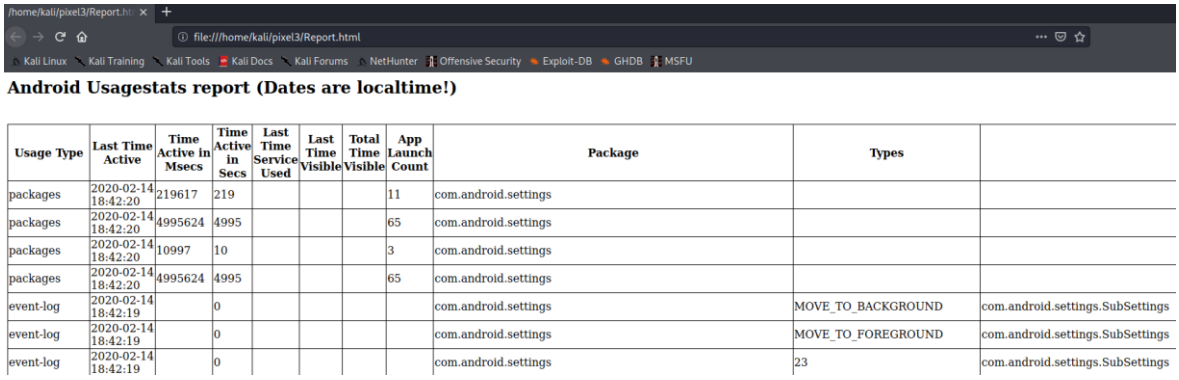


Figure 9. *Manual check: date -d @1580317477 -u returns 29 Jan 2020 17:04:37 UTC for the WeChat first_download_ms value in localappstate.db.*



Usage Type	Last Time Active	Time Active in Msecs	Time Active in Secs	Last Time Service Used	Last Time Visible	Total Time Visible	App Launch Count	Package	Types	
packages	2020-02-14 18:42:20	219617	219				11	com.android.settings		
packages	2020-02-14 18:42:20	4995624	4995				65	com.android.settings		
packages	2020-02-14 18:42:20	10997	10				3	com.android.settings		
packages	2020-02-14 18:42:20	4995624	4995				65	com.android.settings		
event-log	2020-02-14 18:42:19		0					com.android.settings	MOVE_TO_BACKGROUND	com.android.settings.SubSettings
event-log	2020-02-14 18:42:19		0					com.android.settings	MOVE_TO_FOREGROUND	com.android.settings.SubSettings
event-log	2020-02-14 18:42:19		0					com.android.settings	23	com.android.settings.SubSettings

Figure 10. *Output of the usagestats parser (Report.html). The page header states that dates are local time of the examination machine, not device time.*

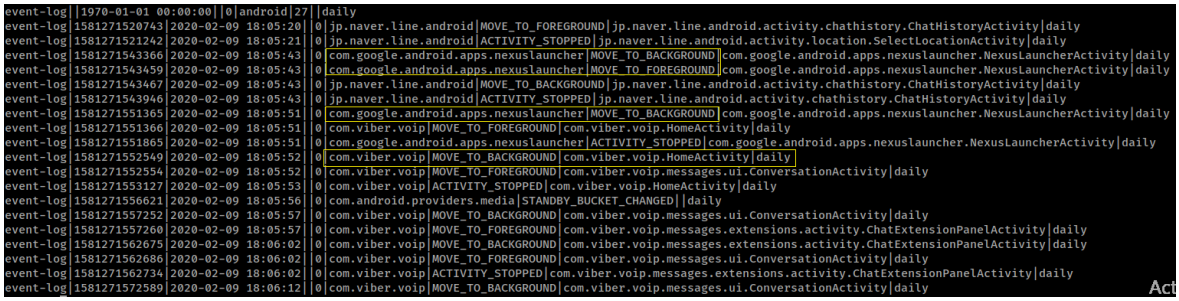
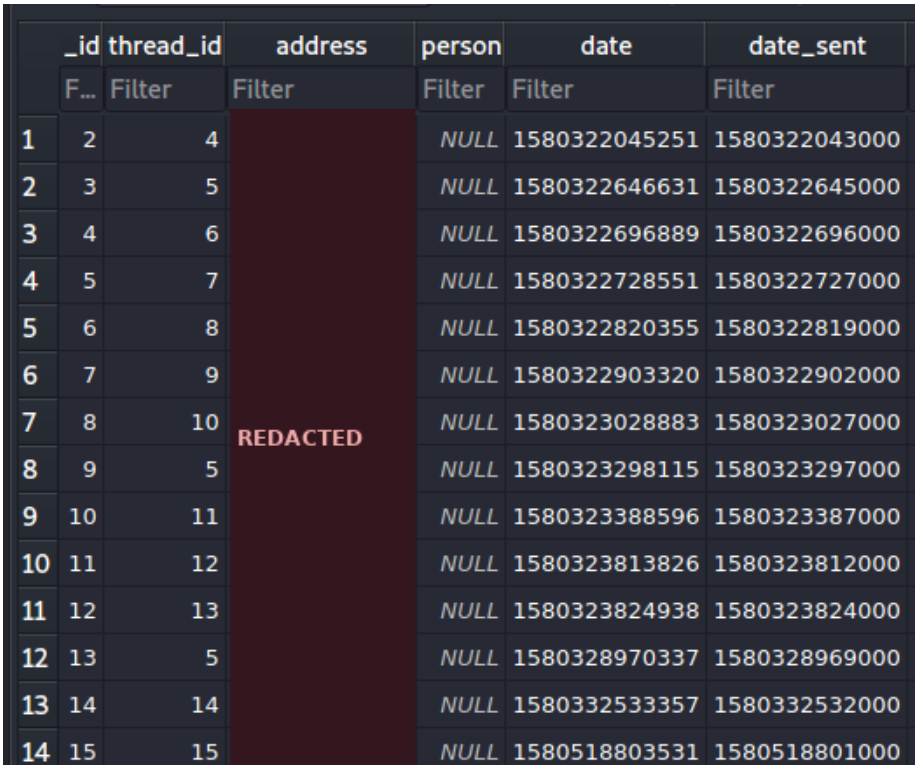


Figure 11. *usagestats.db (parser output) queried with the UNIXEPOCH modifier: LINE and Viber foreground and background events on 9 Feb 2020, raw millisecond values beside UTC conversions.*



	_id	thread_id	address	person	date	date_sent
	F...	Filter	Filter	Filter	Filter	Filter
1	2	4	REDACTED	NULL	1580322045251	1580322043000
2	3	5		NULL	1580322646631	1580322645000
3	4	6		NULL	1580322696889	1580322696000
4	5	7		NULL	1580322728551	1580322727000
5	6	8		NULL	1580322820355	1580322819000
6	7	9		NULL	1580322903320	1580322902000
7	8	10		NULL	1580323028883	1580323027000
8	9	5		NULL	1580323298115	1580323297000
9	10	11		NULL	1580323388596	1580323387000
10	11	12		NULL	1580323813826	1580323812000
11	12	13		NULL	1580323824938	1580323824000
12	13	5		NULL	1580328970337	1580328969000
13	14	14		NULL	1580332533357	1580332532000
14	15	15		NULL	1580518803531	1580518801000

Figure 12. *mmssms.db, table sms: thread_id, date and date_sent (ms). The address column is redacted.*

body	service_center	locked	sub_id	error_code	creator	seen
Filter	Filter	Filter	Filter	Filter	Filter	Fil...
<#> Please enter 304868 into LINE within...	6502531234	0	1	0	com.google.android.apps.messaging	1
Telegram code: 59389...	6502531234	0	1	0	com.google.android.apps.messaging	1
<#> Your Signal verification code: 303-07...	6502531234	0	1	0	com.google.android.apps.messaging	1
Your imo code is 685614...	6502531234	0	1	0	com.google.android.apps.messaging	1
Your Skout verification code is 2661.	6502531234	0	1	0	com.google.android.apps.messaging	1
Your Skout verification code is 9073.	6502531234	0	1	0	com.google.android.apps.messaging	1
<#> Your WhatsApp code: 362-593...	6502531234	0	1	0	com.google.android.apps.messaging	1
Use 3649 as Microsoft account security code	6502531234	0	1	0	com.google.android.apps.messaging	1
[#][TikTok] 6327 is your verification code...	6502531234	0	1	0	com.google.android.apps.messaging	1
FB-55286 is ThisIs's Facebook confirmation...	6502531234	0	1	0	com.google.android.apps.messaging	1
Confirmed! To edit SMS preferences go to ...	6502531234	0	1	0	com.google.android.apps.messaging	1
<#> Your Imgur verification code is ...	6502531234	0	1	0	com.google.android.apps.messaging	1
369967 is your verification code for Dust - ...	6502531234	0	1	0	com.google.android.apps.messaging	1
Your Wickr Me confirmation code is 8434776	6502531234	0	1	0	com.google.android.apps.messaging	1

Figure 13. *mmssms.db, table sms: message body, service centre, seen and creator (com.google.android.apps.messaging).*

	_id	number	presentation	post_dial_digits	via_number	date	duration	data_usage	type	features	subscription_component_name	subscription_id
	F...	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Fil...	Filter	Filter	Filter
1	1	REDACTED	2			1580483796007	0	NULL	3	0	com.android.phone/...	REDACTED
2	2		2			1580494070836	0	NULL	3	0	com.android.phone/...	
3	3		2			1580502115252	0	NULL	3	0	com.android.phone/...	
4	4		2			1580585280414	0	NULL	3	0	com.android.phone/...	
5	5		2			1580758342425	0	NULL	3	0	com.android.phone/...	
6	6		1			1581192896305	0	NULL	3	0	com.android.phone/...	
7	7		1			1581521849939	0	NULL	3	0	com.android.phone/...	
8	8		1			1581521901000	3	NULL	4	0	NULL	
9	9		1			1581555164000	0	NULL	3	0	com.android.phone/...	
10	10		1			1581555213000	3	NULL	4	0	NULL	
11	11		1			1581643741051	11	NULL	2	0	com.android.phone/...	

Figure 14. *calllog.db, table calls (part 1): date (ms), duration, type and presentation. Numbers and subscription IDs are redacted.*

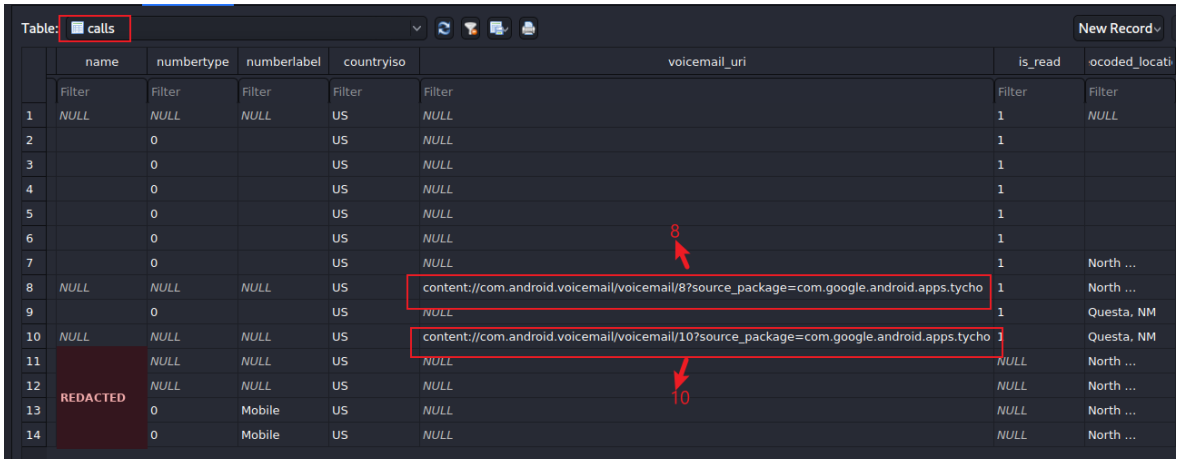


Figure 15. *calllog.db, table calls (part 2): voicemail_uri for rows 8 and 10 (red annotations are the examiner's). Contact display names are redacted.*

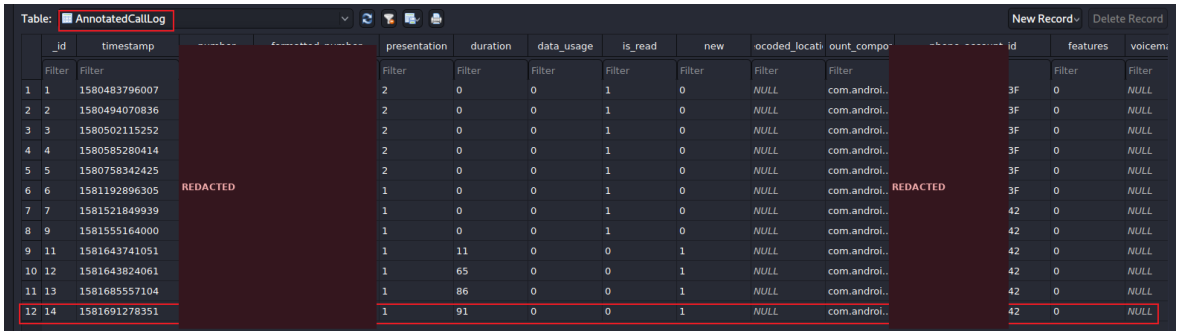


Figure 16. *Google Dialer annotated_call_log.db, table AnnotatedCallLog: timestamp, duration and call identifiers. Rows with _id 8 and 10 are absent. Numbers and account IDs are redacted.*

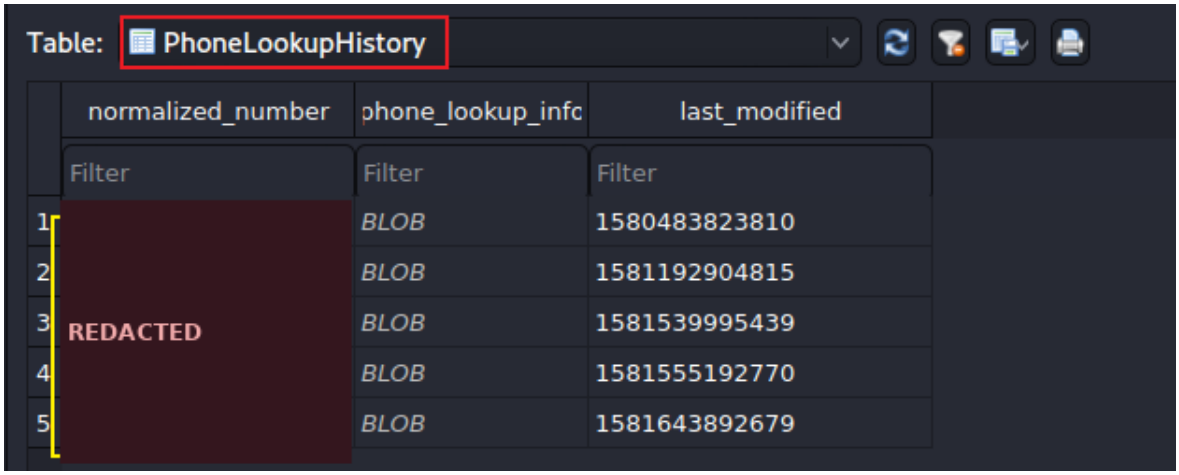


Figure 17. *Google Dialer phone_lookup_history.db: last_modified (ms) for five looked-up numbers (numbers redacted).*

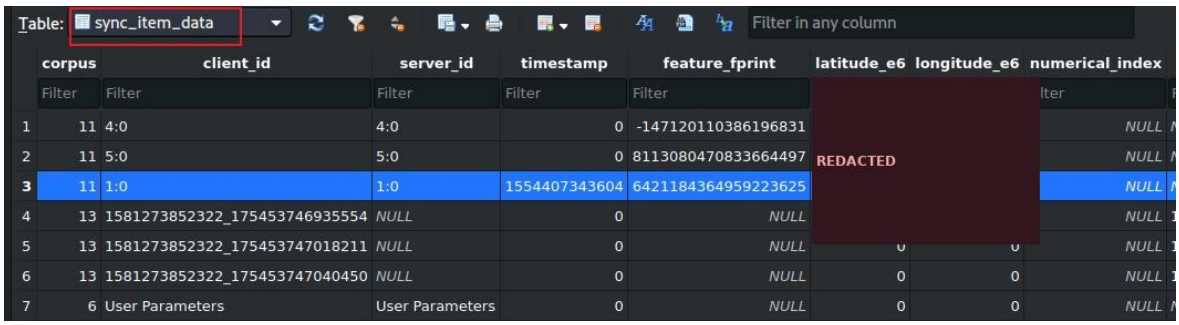


Figure 18. *Google Maps gmm_sync.db, table sync_item_data: latitude_e6 and longitude_e6 columns (E6 microdegrees) are redacted; corpus, client_id and timestamp remain.*

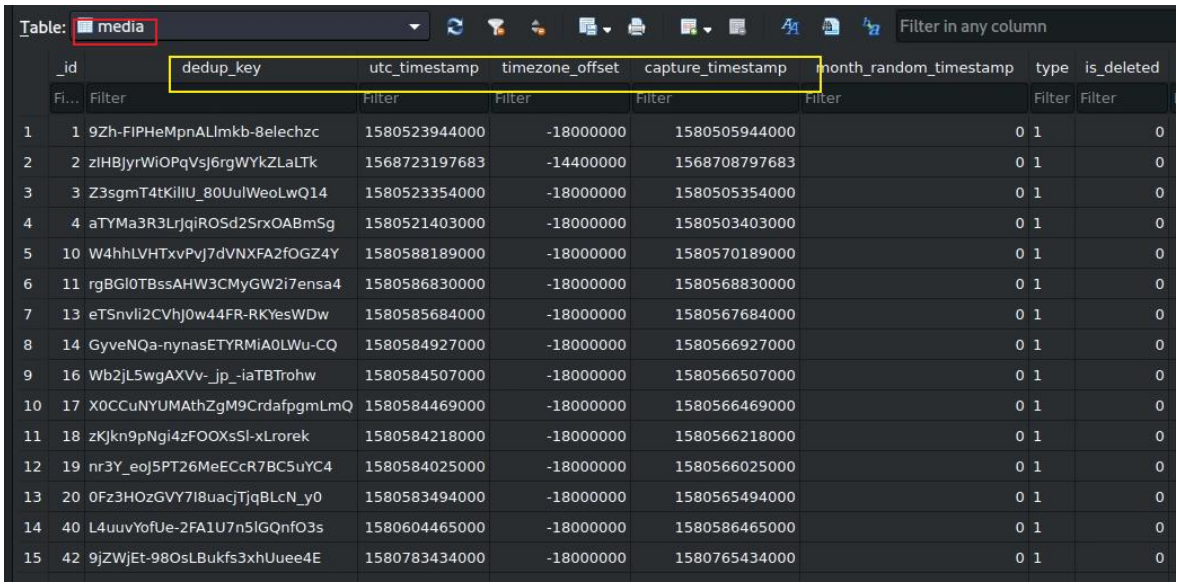
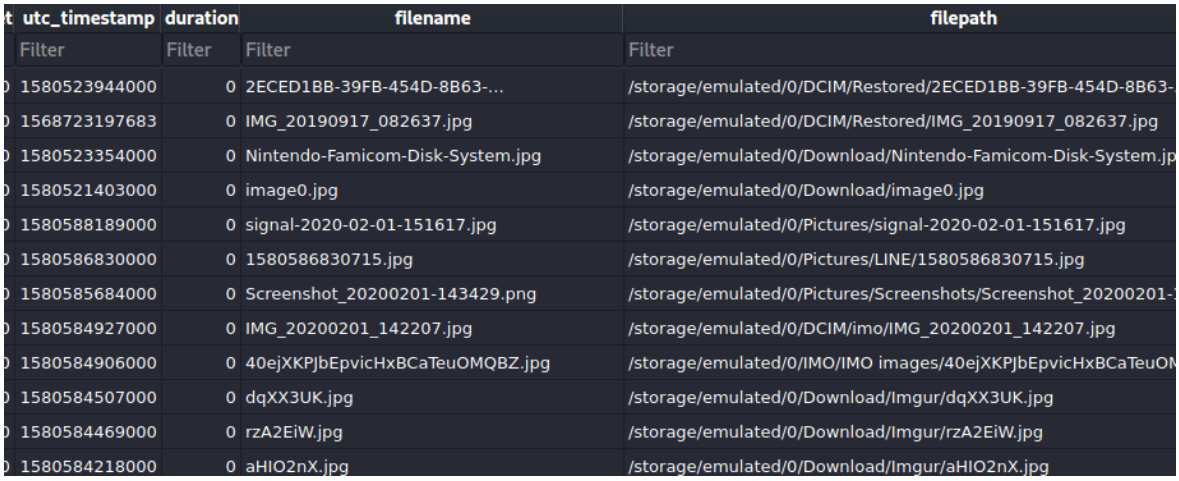


Table: media — Filter in any column

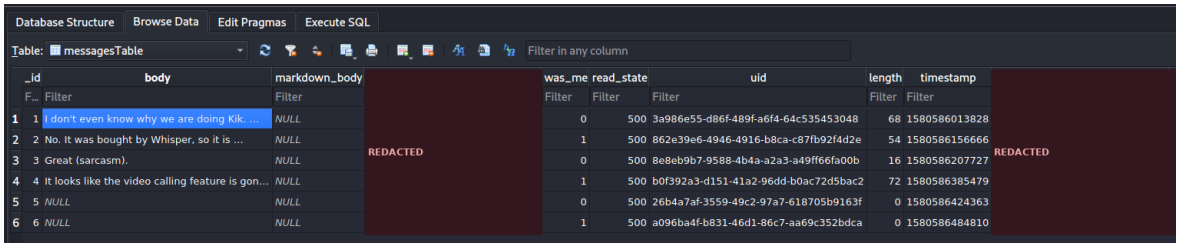
	_id	dedup_key	utc_timestamp	timezone_offset	capture_timestamp	month_random_timestamp	type	is_deleted
1	1	9Zh-FIPHeMpnALlmkb-8elechzc	1580523944000	-18000000	1580505944000	0	1	0
2	2	zIHBJyrWiOPqVsJ6rgWYkZLaLTk	1568723197683	-14400000	1568708797683	0	1	0
3	3	Z3sgmT4tKilIU_80UulWeoLwQ14	1580523354000	-18000000	1580505354000	0	1	0
4	4	aTYMa3R3LrJqiROSd2SrxOABmSg	1580521403000	-18000000	1580503403000	0	1	0
5	10	W4hhLVHTxvPvJ7dVNXFA2fOGZ4Y	1580588189000	-18000000	1580570189000	0	1	0
6	11	rgBGl0TBssAHW3CMyGW2i7ensa4	1580586830000	-18000000	1580568830000	0	1	0
7	13	eTSnvli2CVhJ0w44FR-RKYesWDw	1580585684000	-18000000	1580567684000	0	1	0
8	14	GyveNQa-nynasETYRMiA0LWu-CQ	1580584927000	-18000000	1580566927000	0	1	0
9	16	Wb2jL5wgAXVv-_jp_-iaTBTrohw	1580584507000	-18000000	1580566507000	0	1	0
10	17	X0CCuNYUMAthZgM9CrdafpgmLmQ	1580584469000	-18000000	1580566469000	0	1	0
11	18	zKJkn9pNgi4zFOOXsSl-xLrorek	1580584218000	-18000000	1580566218000	0	1	0
12	19	nr3Y_eoJ5PT26MeECcR7BC5uYC4	1580584025000	-18000000	1580566025000	0	1	0
13	20	0Fz3HOzGVY7I8uacjTjqBLcN_y0	1580583494000	-18000000	1580565494000	0	1	0
14	40	L4uuvYofUe-2FA1U7n5lGQnfO3s	1580604465000	-18000000	1580586465000	0	1	0
15	42	9jZWjEt-98OsLBukfs3xhUuee4E	1580783434000	-18000000	1580765434000	0	1	0

Figure 19. *Google Photos gphotos-1.db, table media: utc_timestamp, timezone_offset (ms) and capture_timestamp.*



t	utc_timestamp	duration	filename	filepath
	Filter	Filter	Filter	Filter
0	1580523944000	0	2ECED1BB-39FB-454D-8B63-...	/storage/emulated/0/DCIM/Restored/2ECED1BB-39FB-454D-8B63-
0	1568723197683	0	IMG_20190917_082637.jpg	/storage/emulated/0/DCIM/Restored/IMG_20190917_082637.jpg
0	1580523354000	0	Nintendo-Famicom-Disk-System.jpg	/storage/emulated/0/Download/Nintendo-Famicom-Disk-System.jp
0	1580521403000	0	image0.jpg	/storage/emulated/0/Download/image0.jpg
0	1580588189000	0	signal-2020-02-01-151617.jpg	/storage/emulated/0/Pictures/signal-2020-02-01-151617.jpg
0	1580586830000	0	1580586830715.jpg	/storage/emulated/0/Pictures/LINE/1580586830715.jpg
0	1580585684000	0	Screenshot_20200201-143429.png	/storage/emulated/0/Pictures/Screenshots/Screenshot_20200201-
0	1580584927000	0	IMG_20200201_142207.jpg	/storage/emulated/0/DCIM/imo/IMG_20200201_142207.jpg
0	1580584906000	0	40ejXKPJbEpvicHxBCaTeuOMQBZ.jpg	/storage/emulated/0/IMO/IMO images/40ejXKPJbEpvicHxBCaTeuOM
0	1580584507000	0	dqXX3UK.jpg	/storage/emulated/0/Download/Imgur/dqXX3UK.jpg
0	1580584469000	0	rzA2EiW.jpg	/storage/emulated/0/Download/Imgur/rzA2EiW.jpg
0	1580584218000	0	aHIO2nX.jpg	/storage/emulated/0/Download/Imgur/aHIO2nX.jpg

Figure 20. *Google Photos gphotos-1.db, table local_media: file names and paths with utc_timestamp (ms).*



Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: messagesTable — Filter in any column

	_id	body	markdown_body	partner_jid	was_me	read_state	uid	length	timestamp	bin_id
1	1	I don't even know why we are doing Kik, ...	NULL	REDACTED	0	500	3a986e55-d86f-489f-a6f4-64c535453048	68	1580586013828	REDACTED
2	2	No. It was bought by Whisper, so it is ...	NULL		1	500	862e39e6-4946-4916-b8ca-c87fb92f4d2e	54	1580586156666	
3	3	Great (sarcasm).	NULL		0	500	8e8eb9b7-9588-4b4a-a2a3-a49ff66fa00b	16	1580586207727	
4	4	It looks like the video calling feature is gon...	NULL		1	500	b0f392a3-d151-41a2-96dd-b0ac72d5bac2	72	1580586385479	
5	5	NULL	NULL		0	500	26b4a7af-3559-49c2-97a7-618705b9163f	0	1580586424363	
6	6	NULL	NULL		1	500	a096ba4f-b831-46d1-86c7-aa69c352bdca	0	1580586484810	

Figure 21. *Kik kikDatabase.db, table messagesTable. The partner_jid and bin_id columns are redacted; was_me 0 or 1 gives direction and timestamp is in ms.*

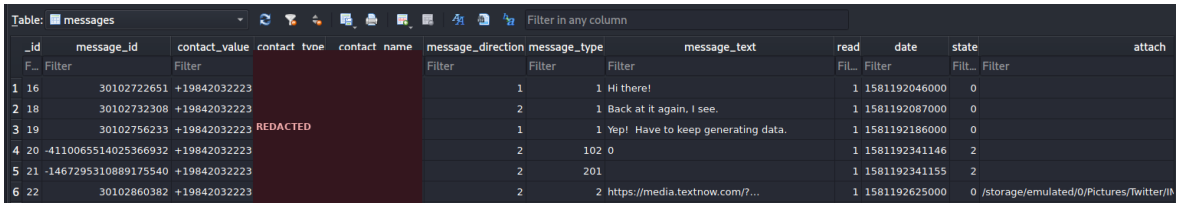


Table: messages — Filter in any column

	_id	message_id	contact_value	contact_type	contact_name	message_direction	message_type	message_text	read	date	state	attach
1	16	30102722651	+19842032223	REDACTED		1	1	Hi there!	1	1581192046000	0	
2	18	30102732308	+19842032223			2	1	Back at it again, I see.	1	1581192087000	0	
3	19	30102756233	+19842032223			1	1	Yep! Have to keep generating data.	1	1581192186000	0	
4	20	-4110065514025366932	+19842032223			2	102	0	1	1581192341146	2	
5	21	-1467295310889175540	+19842032223			2	201		1	1581192341155	2	
6	22	30102860382	+19842032223			2	2	https://media.textnow.com/?...	1	1581192625000	0	/storage/emulated/0/Pictures/Twitter/I

Figure 22. *TextNow textnow_data.db, table messages. Contact columns are redacted; message_direction, message_text and date (ms) remain.*

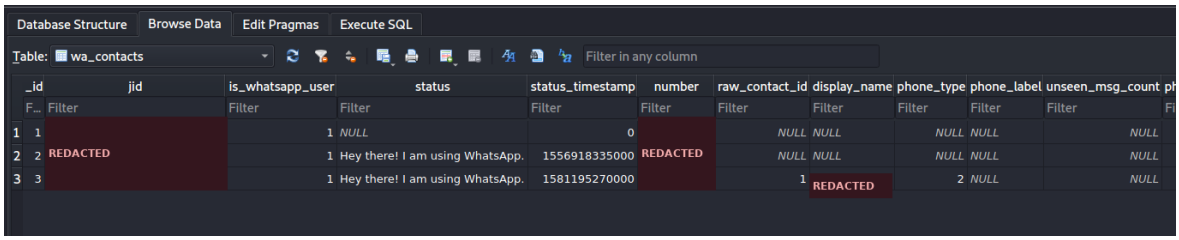


Figure 23. *WhatsApp wa.db, table wa_contacts. JID, number and display name are redacted; status_timestamp (ms) remains.*

Appendix C Queries and Commands

Commands and queries visible in the exhibits are reproduced below.

Purpose	Command or query
Retrieve and hash	wget -q <Digital Corpora Pixel 3.zip URL>; hashdeep "Pixel 3.zip"
Build and patch	cat 'Pixel 3/product/build.prop'; grep patch 'Pixel 3/system/build.prop'
Serial search	grep -ri 'serial number' 'Pixel 3/mnt/vendor/'
Account times (UTC)	select name, type, datetime(last_password_entry_time_millis_epoch/1000,'UNIXEPOCH') from accounts;
Play Store install data	select package_name, delivery_data_timestamp_ms, first_download_ms, account, title from appstate; (exported to appstate.csv)
Manual time check	date -d '@1580317477' -u -> Wed 29 Jan 2020 05:04:37 PM UTC
Ownership time	select purchase_time/1000 from ownership where doc_id like '%whatsapp%' (library.db) -> Thu 29 Nov 2018 07:07:47 PM UTC
Usage parser	git clone Android-Usagestats-XML-Protobuf; python3 usagestats_conv.py (creates usagestats.db and Report.html)
Usage events (UTC)	select usage_type, lastime, datetime(lastime/1000,'UNIXEPOCH'), timeactive, package, CASE types WHEN '1' THEN 'MOVE_TO_FOREGROUND' WHEN '2' THEN 'MOVE_TO_BACKGROUND' ... END, classs, source from data where package='jp.naver.line.android' or package='com.viber.voip'
Name look-up	ls 'Pixel 3/data/data' | while read package; do curl --no-progress-meter https://play.google.com/store/apps/details?id=$package | grep -Pio 'itemprop="name">\K.*?(?=)' && echo $package; done
E-mail address count	grep -EiorhI '([[:alnum:]_.-]+@[[:alnum:]_.-]+?\.[[:alpha:].]{2,6})' 'Pixel 3' | sort | uniq -c | sort -nr

Appendix D Source Paths

Paths are relative to the extracted folder "Pixel 3".

Artefact family	Path	Examined?
Build	product/build.prop; system/build.prop; mnt/vendor/persist	Yes
Users and accounts	data/system/users/userlist.xml, 0.xml, 11.xml; data/system_ce/0/accounts_ce.db; data/system_de/0/accounts_de.db	Yes
SIM, carrier, blocking	data/user_de/0/com.android.providers.telephony/databases/telephony.db; data/user_de/0/com.android.phone; data/user_de/0/com.android.providers.blockednumber	Yes
Connectivity	data/system/users/0/settings_secure.xml, settings_global.xml, settings_system.xml; data/misc/wifi/WifiConfigStore.xml	Yes
Settings and policy	data/system/locksettings.db; notification_policy.xml; netpolicy.xml; gsf and gms databases and shared_prefs	Yes
Packages and Play Store	data/system/packages.list; packages.xml; data/data/com.android.vending/databases/localappstate.db, library.db	Yes
Usage	data/system/package-usage.list; data/system/usagestats/0 and /11	Yes (parser)
Telephony and contacts	data/data/com.android.providers.telephony/databases/mmssms.db; com.android.providers.contacts/databases/contacts2.db, calllog.db, profile.db	Yes
Calendar	data/data/com.android.providers.calendar/databases/calendar.db	Structure only
Google Messages and Dialer	data/data/com.google.android.apps.messaging/databases/bugle_db; com.google.android.dialer/databases/annotated_call_log.db, phone_lookup_history.db, suggest_contact_database, voicemail.db	Yes
Maps and Photos	data/data/com.google.android.apps.maps/databases and files; com.google.android.apps.photos/databases/gphotos-1.db	Partly
Third party	data/data/kik.android; com.enflick.android.TextNow; com.whatsapp; com.snapchat.android	Partly

Appendix E Manual Validation

Three comparisons were made between tool or script output and a source or independent conversion.

Check	Tool output	Manual result	Outcome
Play Store timestamp	localappstate.db first_download_ms 1580317477586 (WeChat)	date -d @1580317477 -u gives 2020-01-29 17:04:37 UTC (Figure 9)	Matched
Ownership timestamp	library.db purchase_time 1543518467665	date command gives 2018-11-29 19:07:47 UTC; an independent conversion agrees	Matched
Usagestats parser	LINE event raw 1581271520743 converted to 2020-02-09 18:05:20 UTC (query); Report.html uses examination-machine local time	Arithmetic conversion agrees. The same event was not read from the original protobuf or XML under data/system/usagestats	Partly verified